



Monitoring petroleum fuel adulteration: A review of analytical methods



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ABSTRACT

Detection of adulteration in the petroleum fuels (gasoline and diesel) is challenging because the adulterants usually include those compounds which are already present in these fuels. Measurements of various physico-chemical properties are employed in addition to chromatography and spectroscopy for determining the compositional variations of these fuels. It is required to ensure their compliance with the regulatory specifications. The inclusion of statistical design and data treatments certainly provide enough sensitivity to discriminate adulterated and unadulterated samples in the case of gasoline. On the contrary, discrimination of diesel from its adulterants is difficult which demands rigorous analytical strategies and unequivocal data interpretation. Through this review, we have attempted to organize the literature information on this subject with an emphasis to identify and optimize methods suitable for monitoring adulteration. Detailed discussions on the role of physico-chemical, chromatographic and spectroscopic methods in gasoline and diesel characterization have been presented.

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1. Introduction

Petroleum is a fossilized mass present below the earth's surface as geological formations. Its raw form is known as 'crude oil' that primarily comprises hydrocarbons of varying molecular weights (MW) and other organic compounds. Its elemental composition is typically distributed among carbon (85–90%), hydrogen (10–14%), sulfur (0.2–3%), nitrogen (<0.1–2%), oxygen (1–1.5%) and organo-metallic compounds (nickel, vanadium, arsenic, lead, and other trace metals) [1,2]. Crude oils are also referred as light, medium or heavy depending on American Petroleum Institute (API) gravity which along with sulfur content governs their price in the international market. Light, medium and heavy crude oils generally have API gravity of greater than 40, within 15–40 and less than 15 respectively [1,3]. The prominent hydrocarbons groups present in the crude oil are paraffins (alkanes), naphthenes (cycloalkanes) and aromatics whereas olefins (alkenes) are formed during the processing of crude. The petroleum processing industry is divided into three distinct sectors i.e. (a) upstream (b) midstream and (c) downstream. The upstream sector carries out exploration of the

crude oil while its transportation from the site of exploration to the refinery is looked after by the midstream sector. Refining of the crude oil into various products including petrochemicals is an important activity in the downstream sector. The petroleum refinery produces different types of hydrocarbon fuels and other valued components through several processes. A variety of finished products such as fuel gas, liquefied petroleum gas, gasoline, aviation fuel, kerosene, diesel, lubricants, waxes, and asphalt etc. are obtained [1,2]. Modern petroleum refineries are equipped with a separate petrochemical complex and important precursors such as ethylene and propylene are formed during various cracking and reforming processes. Aromatics and these precursors serve as starting material for various industries such as solvents, detergents, adhesives, paints, polymer, rubber, dyes etc. [2].

Hydrocarbon fuels from all the refining streams before being supplied to the market for their targeted end-use are required to comply with certain regulatory requirements which are in agreement with the prevailing global norms. Motor spirit (or gasoline or petrol) and diesel are automotive fuels extensively used worldwide for transportation. Being a mixture of low boiling hydrocarbons, gasoline is suitable for spark ignition (SI) type engines where an electrical spark ignites the mixture of air and fuel vapors [2]. Gasoline is a petroleum-derived product comprising a mixture of liquid aliphatic and aromatic hydrocarbons, ranging between C₄ and C₁₂

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carbon atoms with the boiling range of 30–225°C. It is predominantly a mixture of paraffins, naphthenes, aromatics and olefins. Some additives such as aliphatic alcohols and methylethers are also desirable for improving its octane number [4–11]. The composition of gasoline varies as the nature of the crude oil from which it is obtained [12,13]. Diesel fuel is widely used across various sectors such as in the agricultural equipments, automobiles and public transport, household appliances etc. Diesel is combusted in a compression-ignition (CI) engine where the fuel is atomized and sprayed into the hot compressed air [2]. The hydrocarbons types (or classes) in the diesel fuel is typically similar to the gasoline though the MW and carbon number are higher. The usual carbon number range of diesel fuel is within C₁₀–C₁₉ whilst the boiling range lies within 180–370°C. Similar to the gasoline, variation in the nature and composition of diesel fuel is generally accounted due to the difference in the nature of crude oil and refining processes [14].

The energy requirements are increasing at a rapid pace in the developing world putting a thrust on the consumption of fossil fuels. International Energy Agency has accounted for, in general, the global energy supplies, gross consumption across various sectors and anticipated requirements in the future [15]. Worldwide, the transportation is considered a significant energy demanding sector, requirements of which are largely met through fossil fuels. Besides, the Indian transport system expanding at 10% annually is considered one of the largest and heavily populated systems with serious fuel consumptions [16,17].

1.1. Fuel adulteration

Adulteration is an illegal introduction of any foreign substance into any product leading to the non-compliance with its standard specification. In the developing countries, it is rampant in the automotive fuels (gasoline and diesel) segment and causes deterioration in the fuel quality. It is detrimental to the engine's optimum performance and may result in damaging its parts. Adulteration also augments vehicular tailpipe emissions which are considered hazardous to the environment and living beings. In addition, it also causes a negative impact on the tax policy of Governments and economic development. Adulteration of petroleum-based automotive fuels is a serious concern globally and a menace in the developing economies. The fuel quality is an important parameter for the environmental assessment and a measure of public health, essential for the developing economies [18]. Fuel adulterants are selected on the basis of several criteria keeping in mind the specifications of the final product. For financial benefits, the substantial difference between the pricing of adulterant and fuel is considered. Availability of cheaper supplies (such as industrial solvents and other low-value streams) resembling the finished fuel properties is a profitable scenario. In most of the cases, chemicals and solvents originating from petroleum refining processes satisfy the requirements and used as the adulterant for gasoline or diesel fuel.

Gasoline is commonly adulterated with naphtha, kerosene, diesel and other solvents available at lower prices for industrial end-use only. In India, kerosene distributed through public distribution system (PDS) is a subsidized product and is provided to economically weaker sections by the Government. The PDS kerosene is particularly utilized for adulteration purpose along with naphtha, rubber solvents (pentanes and hexanes), aromatics (benzene, toluene, and xylenes), ethanol, light (C₄–C₈) and heavy aliphatic hydrocarbons (C₁₃–C₁₅), thinners (toluene, xylenes, acetates and alcohols), white spirits, turpentine and used lubricants [7,19–22]. Diesel fuel is often adulterated with cheaply available domestic kerosene fuel [23]. Kerosene has strong potential for being used as an adulterant in the diesel due to overlapping hydrocarbons (C₉–C₁₉ in diesel and C₆–C₁₆ in kerosene). In addition, bio-

diesels obtained from edible and non-edible seeds (soybean, corn, cotton, palm oil etc.) and residual oils (automotive lubricant oil and used frying oil etc.) from several processes have also been reported as adulterants of diesel [24–26].

1.2. Environmental concerns

Exhaust emissions from vehicles are a prominent source of harmful pollutants that include toxic gases and particles largely associated with the known health hazards due to inhalation exposures in living beings. Gases typically include water vapor and oxides of carbon (CO, CO₂), nitrogen (NO, N₂O, NO₂) and sulfur (SO₂) [27]. Hydrocarbons form an important constituent of the vehicular exhausts and arise due to incomplete combustion of fuel inside the engines. Mechanistically, hydrocarbons which enter the engine during fuel intake get partitioned in the lubricant film over cylinder wall. This portion of non-combusted fuel components is vented out of the engine during expansion and exhaust stroke of the combustion cycle. The exhausts also contain fine particulate matter that acts as a carrier to several toxic organic chemicals such as 1,3-butadiene, aldehydes (formaldehyde, acetaldehyde, acrolein etc.), benzene, polycyclic aromatic hydrocarbons (PAHs) and nitrated-PAHs [28–30].

Environmental assessment has attracted the attention of environmental scientists and tremendous research has been witnessed in this area. Mitigation of the environmental concerns due to worsening air pollution and emission of greenhouse gases have resulted in the enforcement of emission standards for the transportation sector. Bharat stage (BS) emission norms are regulatory standards enforced by the Government of India since 2000 and based on European regulations. The norms have progressed stringently in view of the rising air pollution levels and have helped in putting a check on it. It had resulted in the increase in vehicle costs necessitated due to improvement in technology and fuel prices also increased. The vehicle manufacturers were required to enforce these regulatory measures and comply with the allowed emissions. Since 2010, BS III norms were endorsed across the country while BS IV emission norms were also effective in the major cities. In 2016, the Government declared that BS V norms would be skipped and BS VI norms will be straightaway implemented by the year 2020. Despite all the efforts, the air quality in India is still ranked poorly at the 141th place as per Environmental performance index 2016 whereas it was at 125th place 4 years back [31]. Various factors such as traffic congestion, lack of proper roads, inadequate traffic laws and in particular fuel adulteration have contributed in worsening the air quality. The World Bank stated in 2013 that air pollution and other environmental degradation costs India \$80 billion per year, which is about 5.7% of the country's GDP [32]. Fuel adulteration is a significant contributing factor towards the deteriorating air quality which has been noticed by authorities including Supreme Court of India [33].

Catalytic converters (or particulate filters) are used with either gasoline or diesel powered internal combustion engines and control the emissions. It catalyzes redox reactions and converts toxic constituents of the emissions into less toxic gases. Regular regenerations of these catalytic converters at high temperature are required to remove the trapped soot. Toxic emissions in the form of hydrocarbons such as benzene and PAHs depend mostly on the fuel composition and catalyst performance. Usually, the kerosene has a boiling point range of 145°C–300°C, which is higher than the boiling range of gasoline. Consequently, if the gasoline is adulterated with kerosene, it is more difficult to burn the kerosene inside the engine, leading to an increase in the hydrocarbons, CO, and particulate matter emissions. If the kerosene adulteration is too much, the octane quality falls below the octane requirement of the

engines and engine knocking occurs. Alternatively, in a scenario where gasoline is adulterated with diesel at lower levels, the same general effects are observed, however, if the adulteration levels are high, a significant increase in hydrocarbons, CO and particulate matter is observed. Further, if the gasoline has adulterants in the identical boiling range (comprising of industrial solvents), the specifications are usually met and no major compromise in the driveability of the vehicle is observed. If the amount of toluene and xylenes are high, there would be an increased hydrocarbons, CO and NOx emissions in addition to the significantly high levels of benzene [19,34].

As per the Bureau of Indian Standards (BIS) document (IS 2796:2010 and IS 15464), the permissible limit of anhydrous ethanol blend in gasoline is 5% (v/v). As compared to gasoline, ethanol has lesser energy per unit volume as a result of which fuel economy is reduced, however; the presence of ethanol in gasoline is also beneficial since ethanol is a very high octane fuel and act as an octane enhancer. Excessive ethanol concentration (used as adulterant) in gasoline can cause an increase in CO emissions. A consequence of being hygroscopic, ethanol blended gasoline is susceptible to moisture which creates the problem in engine operation, corrosion of the fuel tank and internal engine components [35] (Table 1).

Efficient analytical approaches with valid conclusions are extremely important to tackle this menace. Tools of analytical chemistry offer discrete information on measurement varying from bulk properties to the molecular level. It needs to be highlighted here that standard regulatory norms including test methods which have been adopted by the countries are required to ensure the compliance with the laid down specifications. In this context, testing of finished fuels with standard methods have the objective of assessing the deviation in their quality. In the absence of separate methods for detecting adulteration in fuels, respective standard test methods are practiced to assure the compliance with regulatory norms. Under the prevalent circumstances, availability of various analytical techniques is highly advantageous in detecting the fuel adulteration. It would be beneficial to have organized information about a range of analytical procedures being adopted currently for petroleum fuel characterization and their extended use towards detecting adulterants. Broadly, the characterization is carried out within the two broad domains: physico-chemical and chromatographic/spectroscopic methods and have been reviewed in the ensuing discussion (Table 2).

2. Gasoline and diesel: analysis and detection of adulteration

Complexity in the composition of petroleum-derived fuels and their potential adulterants leads to a challenging scenario where enforcement and compliance of standard norms are often not easy. Owing to its diverse implications, research in monitoring the fuel adulteration was extensively undertaken by a large number of

researchers. Widely accepted and adopted standard methods include ISO, EN, and ASTM that encompass a variety of properties for testing of petroleum products including fuels. As already commented, no test is specifically designed to measure the adulteration of petroleum products although, most of these methods are equally applicable to monitor adulteration of gasoline and diesel fuels.

2.1. Quality assurance/control approaches

The quality assurance (QA) and control (QC) procedures are absolutely essential in monitoring and ensuring the quality of all components involved in any process/products. The regulatory standards related to the QA/QC of finished petroleum fuels are required to be strictly complied before their intended use. In accordance with the required applicable norms, several standard protocols, non-standard and laboratory developed methods are being used in the industry. In the oil sector, QC of finished diesel, for example, have to adhere to the compliance with EN 590/ASTM D975 for which different countries have adopted them. The major criteria as per this standard involve Cetane number determination by ISO 5165, EN 15195, EN 16444 and ASTM D613. In addition, ISO 4264/ASTM D976 is needed for Cetane index. Similarly, specific tests related to density (ISO 3675/ISO 12185), aromaticity/PAHs (EN 12916, ASTM D1319), sulfur content (ISO 20846), flash point (ISO 2719/ASTM D93), kinematic viscosity (ASTM D445/ISO 3104) and various others have been listed and discussed. Intermediate quality checks are carried out onsite (at filling station) for QC of gasoline and diesel through density measurement by Hydrometer method (ASTM D1298). Needless to mention that measurement traceability of all the equipment(s) and apparatus is linked through the unbroken chain with the international metrological standards. In the same way, offsite (at laboratory) measurement of density is widely practiced by digital density meter method (ASTM D4052) for gasoline and diesel.

In the area of petroleum testing, validation of the test methods is pertinent to the scope and outcome of any analysis. All the standard methods have been developed after rigorous inter-laboratory comparisons and proficiency testing programs and considered milestones in the analysis of gasoline and diesel fuels. Deviations in the test results are assumed to arise in situations indicating a slight or significant change in the composition. Adulteration is one such manifestation, which may be considered a root cause of such variations in the composition. An outline of the important methods (standards) used in gasoline and diesel QC is provided in Fig. 1. The prevalent Indian standard (BS-IV) corresponding to its counterpart (Euro-IV) for QA/QC of gasoline (IS 2796:2010) and diesel (IS 1460:2010) is also provided as supplementary information. With the use of permissible amounts of ethanol in gasoline, RON/MON estimation using ASTM D2699 and ASTM D2700 tolerates the presence of ethanol by up to 25%. ASTM D5580 is a standard method which is used for the determination of total aromatics in finished gasoline

Table 1
Typical distillation and carbon number range of petroleum products.

S. No.	Petroleum distillates	Boiling range/Carbon range
1.	Liquefied petroleum gas	Up to 30°C (C ₁ –C ₄)
2.	Gasoline	30°C–225°C (C ₅ –C ₁₁ or C ₁₂)
3.	Aviation fuel	150°C–300°C (C ₁₀ –C ₁₄)
4.	Kerosene	140°C–320°C (C ₉ –C ₁₆)
5.	Diesel	180°C–380°C (C ₁₁ –C ₂₀)
6.	Fuel oil	216–421°C (C ₁₂ –C ₂₀ or above)
7.	Lubricating oil	More than 400°C (C ₂₅ –C ₄₀ , some times C ₂₅ –C ₅₀)
8.	Wax	More than 300°C (>C ₁₇)
9.	Asphalt	More than 340°C (>C ₂₀)

Table 2
Approaches of analytical chemistry for detection of adulteration in petroleum fuels.

S. No.	Analytical approach	Parameters	Gasoline	Diesel	Test method	References
1.	Physico-Chemical properties	Density	✓	✓	ASTM D1298/ASTM D4052	[12,14,39,44]
		Distillation	✓	✓	ASTM D86	[12,13,20,35,39]
		Research octane number	✓	×	ASTM D2699	[35,39]
		Motor octane number	✓	×	ASTM D2700	[35,39]
		Filter paper method	✓	×	—	[13]
		Elemental content	✓	✓	ASTM D2622/ASTM D5453/ASTM D5059/IP 352/IP 224	—
		Vapor pressure	✓	×	ASTM D5191/ASTM D6378	[12]
		Kinematic viscosity	×	✓	ASTM D445	[14,44,45]
		Cetane number	×	✓	ASTM D613	—
		Cetane index	×	✓	ASTM D4737	—
		Flash point	×	✓	ASTM D93	—
		Pour point	×	✓	ASTM D5950/ASTM D5985	—
		GC: C ₂ to C ₅ hydrocarbons	✓	×	ASTM D2427	—
		GC: Boiling range distribution	✓	✓	ASTM D887	—
		GC: Benzene and toluene	✓	×	ASTM D3606/ASTM D6370/UOP 744	—
2.	Chromatography	GC: Ethers and C ₁ to C ₄ alcohols	✓	×	ASTM D4815	—
		GC: Hydrocarbon types, oxygenates	✓	×	ASTM D6839, ASTM D6730	[6,13,49]
		GC: BTEX, C ₉ , heavy aromatics and total aromatics	✓	×	ASTM D5580/UOP 744	—
		GC: Oxygenated content	✓	×	ASTM D5599	—
		GC: Sulfur compounds	✓	×	ASTM D5623	—
		GC: Total olefins	✓	×	ASTM D6296	—
		HPLC-reversed phase: Determination of kerosene	✓	×	—	[46]
		HPLC: Quantification of triacylglycerols in diesel	×	✓	—	[25]
		SFC: Aromatic content and PNA	×	✓	ASTM D5186	—
		SFC: Olefin content	✓	×	ASTM D6550	—
		HPLC: Aromatic hydrocarbon types	×	✓	ASTM D6591/IP 391	—
		GC×GC	✓	✓	—	[21,51,52,54,59]
		MS: Hydrocarbon types in petroleum fuels	✓	✓	ASTM D2789	[60]
					ASTM D2425	—
		MS: Benzene, toluene and total aromatics	✓	×	ASTM D5769	—
3.	Spectroscopy	GC–MS: Identification of solvents in gasoline	✓	×	—	[10]
		HS–MS: Adulteration in gasoline samples on the basis of RON	✓	×	—	[9]
		GC–MS: Determination of biodiesel in diesel	×	✓	—	[58,59]
		Fluorescence	✓	✓	—	[62–67]
		Flame emission	✓	✓	—	[68]
					ASTM D7111	—
		NMR: Aromatics	×	✓	—	[23,74]
		NMR: Hydrocarbon types in gasoline	✓	×	—	[5,8,70,71]
		NMR: Hydrogen content	✓	✓	ASTM D4808	—
		GC–FTIR: Oxygenates, benzene, toluene, C ₈ –C ₁₂ aromatics and total aromatics	✓	×	ASTM D5986	[79]
		Infrared: Adulteration monitoring	✓	✓	—	[22,75–79,81,83]
		Raman: Gasoline adulteration	✓	×	—	[82,85,86]
		Photo-thermal	✓	×	—	[34]
		Acidity	×	✓	ASTM D664	—
		Water content	✓	✓	ASTM D95	—
4.	Volumetry	Phase-titration method	✓	✓	—	[87,88]
		Gravimetric technique	✓	×	—	[92]
5.	Gravimetry	Thermogravimetric technique (TG/DTG)	✓	×	—	[91]

including benzene, toluene, ethylbenzene and xylene along with C₉ and heavier aromatics. ASTM D4815 is used for determining total alcohols in gasoline whereas ASTM D6730 is used to resolve individual hydrocarbons in a 100 m capillary column. It gives the quantitative determination of individual hydrocarbons with theoretical relative response factors and also provides information about various oxygenated organics.

2.2. Physico-chemical methods

Density, distillation, research octane number (RON), motor octane number (MON) and Reid vapor pressure are frequently used for detecting adulteration in gasoline and provide primary evidence of its adulteration. In this context, density is estimated per unit volume of the fuel measured at 15°C. Irrespective of the processes from where the gasoline or diesel fuels are obtained, density measurement ensures uniformity in their hydrocarbon distribution

[1,36]. Similarly, the atmospheric distillation of these fuels (as per ASTM D86) in a lab-scale batch distillation unit reveals quantitative determination of boiling range characteristics. Comparison of the boiling range gives information on the composition and properties of components present in pure and adulterated samples [36]. The octane number depends upon the chemical composition of gasoline which falls below the engine's octane requirement in the case of any adulteration [19,36]. In a similar manner, variation in the Reid vapor pressure also indicates contamination by relatively volatile or non-volatile adulterants [36]. Among the early reports, Babitz et al. reported the determination of kerosene adulteration in the motor gasoline [37]. Oliveira et al. described a procedure based on multivariate pattern recognition and gasoline adulteration with ethanol was studied. Input data were provided from distillation curves whilst density and octane number were also measured. The authors claimed that the adopted model furnished optimum results, however; additional confirmation from spectroscopic and

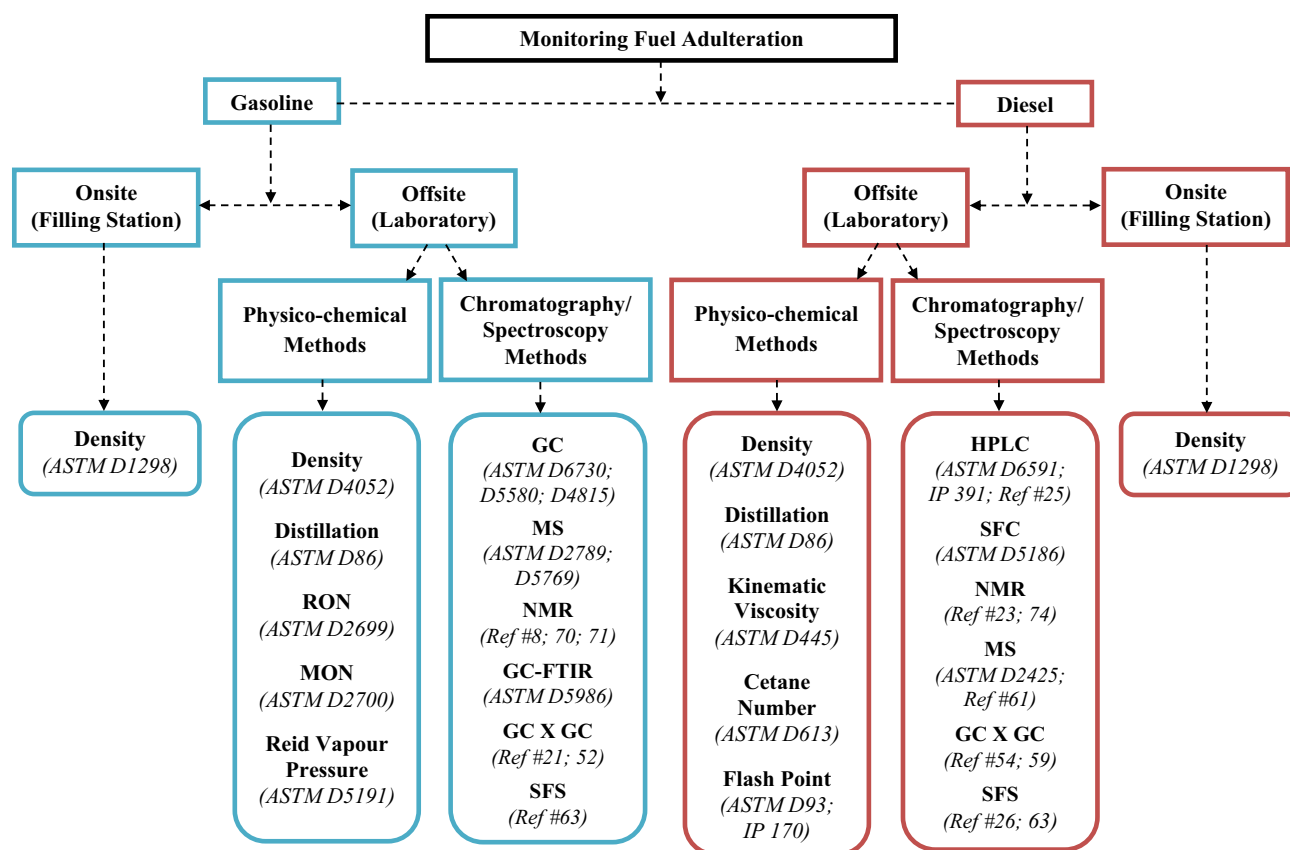


Fig. 1. Important test methods for monitoring fuel adulteration.

chromatographic techniques was emphasized [35]. Evaluation of density, distillation and octane number after varying ethanol concentration in gasoline has been studied [38]. Regular and special Brazilian gasoline samples were analyzed through multivariate statistical technique and physico-chemical methods. The data reportedly served as an important source for determining adulteration in commercial gasoline samples across Brazil [39].

Variation in the octane number, specific gravity, Reid vapor pressure and atmospheric distillation data of premium motor spirit samples in Nigeria concluded significant adulteration when compared with the data collected from unadulterated samples [40]. A very simple method has been demonstrated which showed rapid vaporization of unadulterated gasoline than its adulterated counterpart due to boiling range difference when dropped onto a clean and dry filter paper [13]. A study involving multivariate analysis to distinguish between adulterated and unadulterated gasoline samples with 97% accuracy is reported. The authors have advocated that the developed method had an advantage in terms of the possibility of extending it directly to routine analysis without further validation. However, in our opinion, the external validity generally ensured through round-robin tests is lacking and must be addressed adequately [20]. In a separate study, four adulterants falling into oxygenated, light aliphatics, aromatics and heavy aliphatics classes were selected and influence on Reid vapor pressure, density, and distillation curves were monitored [12].

On the similar pattern, diesel adulteration is usually monitored through determination of important physico-chemical parameters such as density, kinematic viscosity, distillation, cetane number and flash point etc. On the basis of preliminary evidence of adulteration provided by these tests, an eventual detailed component analysis may be performed. Kinematic viscosity indicates the magnitude of

the existing intermolecular forces within a sample which implies that non-polar and polar solvents exhibiting low and high viscosities respectively provide a ground to differentiate between adulterated and unadulterated diesel fuel [41]. Variation in the Cetane number and flash point in the diesel fuel also indicate adulteration since these parameters depend on the chemical composition of samples [36,42,43]. Kerosene being a common adulterant in diesel has been studied extensively by researchers. In a study, authors have concluded that density is not a good indicator of adulteration and highlighted the role of Kinematic viscosity in monitoring the diesel adulteration with varying amounts of kerosene. The results were supplemented with opacity measurement of the emission resulting from these blends indicating an adulteration level ranging between 35 and 50%. The study, however, appears inadequate with peripheral observations devoid of validation procedure [44]. Aleme et al. demonstrated that the distillation curves (ASTM D86) along with multivariate calibration were efficient in predicting specific gravity and kinematic viscosity regardless origin and type of diesel. Similarities in results were also showed for ASTM D4052 and D445 with high repeatability, reproducibility, and confidence [45]. In another study, the kinematic viscosity was selected as a basis for determining adulteration of diesel with kerosene [41].

2.3. Chromatographic methods

With the limited application in the area of monitoring adulteration, high performance liquid chromatography (HPLC) offers a minor role in detecting gasoline adulteration. HPLC methods have been reported employing reversed-phase chromatography for the determination of kerosene as an adulterant in gasoline with ultraviolet (UV) detection. The RSD values for gasoline, kerosene,

synthetic mixtures of gasoline and kerosene were reported in the range of 0.66%–4.36% [46]. However, in monitoring the diesel adulteration, there are few reports concerning HPLC methods on identifying the adulterants. Brandao et al. have reported the detection of adulteration of diesel with vegetable oils through HPLC-based method based on quantification of triacylglycerols. The method utilized reversed-phase column with UV detection [25].

Gas chromatography (GC) being a versatile, simple, efficient and powerful technique has been enormously used for this purpose. The technique reveals molecular level information and the results are unambiguous especially for gasoline. Detailed hydrocarbon analyzers (DHA) are widely used for comprehensive gasoline. The most popular detector employed for this purpose is universal and highly sensitive for the hydrocarbons i.e. flame ionization detector (FID) [47]. Wiedemann et al. reported a multivariate statistical analysis of gasoline samples spiked with adulterants and recorded physico-chemical parameters in addition to exact compositional analysis of the samples through GC [11]. Detection of solvent adulteration in gasoline is difficult because of the possibility of its prior presence, however; their disproportionate existence can be easily monitored through GC-FID [4]. GC-DHA was designed for separation of gasoline compounds accurately up to C9 in a straight run hydrocarbon fractions using single 100 m long capillary column [48]. Detailed component wise analysis of a large number of Brazilian gasoline samples has been reported using GC-FID equipped with DHA with the RSD values for studied analytes below 10% [6]. Similarly, adulteration of motor gasoline with kerosene in India has also been studied and reported using GC-DHA [49].

Despite the availability of a large number of literature reports, evaluation of gasoline adulteration through GC is not a trivial task. Over the past few years, comprehensive two-dimensional gas chromatography (GC×GC) has been shown to possess huge potential for investigating complex mixtures such as petroleum products. It has emerged as an efficient tool owing to its improved resolution and therefore seen as a promising alternative to GC [50]. GC×GC is popularly used for the characterization of petroleum and petroleum derivatives, through group type analysis such as BTEX (benzene, toluene, ethylbenzene, xylenes), total aromatic hydrocarbons, PAHs, heteroatomic compounds (sulfur, oxygen, and nitrogen-containing) in petroleum-derived fuels and other products [48,51]. A detailed characterization of gasoline samples adulterated with kerosene is reported through GC×GC using FID. Similar experiments were also carried out with gasoline samples adulterated with solvents and other adulterants through GC×GC. The authors have also provided good values of the relative errors i.e. ranging between −1.8 and +36.0 obtained with different blending proportions [21]. A multivariate analytical method using GC×GC with FID detector has been proposed to quantify kerosene in type-C Brazilian gasoline samples [50]. Results of GC×GC analysis of aromatic components of gasoline and diesel fuels have been reported and compared with GC–MS in separate studies [52,53]. The R-squared values for benzene, toluene, ethylbenzene, xylene, and other aromatics in gasoline samples were also reported in the range of 0.9986–1.0000 with RSD of less than 2 for each studied compound [52]. Similarly, a GC×GC-FID method has been reported for the analysis of petroleum diesel, biodiesel, and biodiesel + diesel blends under typical conditions [54]. In all the methods relating GC×GC separation, the immense volume of data is generated which is deconvoluted through operating software. Method development involving phase selection, modulation and processing is essential for petroleum fuel analysis. GC×GC methods provide a superior level of information as compared to other modes of chromatography, however; validation of procedures becomes extremely important in view of this. A standard test method (ISO 22854:2016) pertaining to the use of

GC×GC determination of hydrocarbon types and oxygenates in gasoline is available for this purpose (Table 3).

Due to the increased complexity of diesel fraction due to the presence of increased number of isomers of hydrocarbons from various classes, chromatographic methods are of little help in determining the composition and/or adulteration. It is also evident through the presence of the negligible number of literature reports in this field. GC×GC being a powerful tool has been used widely with the mass selective detection (MS) which has been discussed in the consequent discussion.

2.4. Spectroscopic methods

2.4.1. Mass spectrometry

Gas chromatography coupled to mass spectrometry (GC–MS) has also been extensively employed for molecular level information related to petroleum distillates. In the context of adulteration, chemometrics has been utilized for identifying solvents in gasoline in conjunction with GC–MS. The authors claimed to identify the selected solvents at very low concentrations (2%) which have been highlighted as the merit in their study [10]. Although not related directly with the adulteration, a method has been reported for the classification of different gasoline samples on the basis of RON and headspace-MS method. The headspace based extraction has been proposed as an effective quality control method and appears to be a promising approach [9]. An experiment that utilized GC–MS for quantitative estimation of methyl tert-butyl ether and benzene in gasoline is a direct approach for adulteration monitoring [55]. A novel approach for quality control and adulteration check for gasoline, diesel, biodiesel, and other petroleum distillates has been proposed by Alberici et al. The technique has been termed “easy ambient sonic-spray ionization mass spectrometry” is virtually free of sample preparation steps and suitable for hyphenation and onsite analysis [56]. Another application of “easy ambient sonic-spray ionization mass spectrometry” is reported for fingerprinting of gasoline, kerosene and diesel samples. Nitrogen-containing aromatic compounds were detected in their protonated forms and considered ‘markers’ for that sample. The study clearly characterized admixtures of gasoline/kerosene and gasoline/diesel with excellent limits of detection (LOD) of 10 and 1% (v/v), respectively [57].

Determination of the degree of blending bio-diesels in diesel has been investigated with multiple techniques by various groups. Multivariate analysis of the bio-diesel/diesel blends for predicting their composition were carried out with GC–total-ion-current mass spectrometry, GC–MS, GC×GC–total-ion-current mass spectrometry GC×GC–MS. While comparing these analytical approaches the study reported an average error in prediction at around 12% [58]. In a separate study, determination of the origin of biodiesel in a diesel/biodiesel blend by individual quantification of fatty acid esters in kerosene and diesel has been compared through GC×GC-FID and GC×GC–time-of-flight mass spectrometry. The study also investigated group type classification of hydrocarbons with these techniques and comparison with ASTM D2425 has been made. Although little information on the analytical figures of merit was provided, the relative errors of prominent hydrocarbon classes and several methyl esters in diesel were estimated in the range of 0.35%–7.85% with the confidence level of 99% on the basis of five replicates [59]. Bricker et al. had conducted a detailed and well-planned study to compare the results of a modified GC–field ionization mass spectrometry, DHA, ASTM D2786 and ASTM D3239. It estimated detailed hydrocarbon types in the diesel fuel which after careful correlation may be easily extended to monitor the changes in fuel hydrocarbon type profiles from the adulteration point of view [60].

Table 3

An overview of important standard and laboratory developed methods with their advantages and drawbacks.

S. No.	Analytical method	Instrumental technique	Advantage(s)	Drawback(s)
1.	ASTM D5580	GC	Standard method for determination of benzene, toluene, ethylbenzene, xylenes, C ₉ & heavier aromatics and total aromatics in finished motor gasoline. Aromatic hydrocarbons separated without interferences from other hydrocarbons.	Non-aromatic hydrocarbons having b.p. greater than <i>n</i> -dodecane identified as interferents for C ₉ and heavier aromatics. <i>p</i> -xylene and <i>m</i> -xylene unresolved. C ₉ and heavier aromatics are determined as a single group. Applicable to the limited concentration range i.e. 0.1–5.0% for benzene, 1.0–15.0% for toluene, 10.0–80.0% for total aromatics (in vol%).
2.	ASTM D4815	GC	Standard method for the determination of C ₁ to C ₄ alcohols in gasoline.	Applicable to the limited concentration range for individual alcohols i.e. 0.20%–12.0% (mass%).
3.	ASTM D6730	GC	Highly useful standard method for determination of gasoline boiling range adulterants individually including oxygenates within the concentration range of 1.0%–30% (mass%).	Not applicable for detection of hydrocarbons above gasoline (kerosene, diesel etc.).
4.	ASTM D2789	MS	Standard method for determination of hydrocarbon classes in gasoline.	Olefin tolerance is 3% hence not applicable to the finished gasoline from mixed streams.
5.	ASTM D2425	MS	Standard method for determination of hydrocarbon classes in middle distillates including diesel.	Requires prior separation of the sample into saturated and aromatic fractions through ASTM D2549.
6.	ASTM D5769	GC–MS	Standard method for determination of gasoline range industrial aromatic adulterants (benzene, toluene, xylenes and total aromatics) in gasoline.	Applicable to the limited concentration range of 0.1%–4.0% for benzene, 1.0%–13.0% for toluene and 10.0%–42.0% for total aromatics (in vol%).
7.	[8]	NMR	Method for the determination of aromatics, olefins and paraffin including ethanol and benzene in gasoline. Does not require any pre-treatment. Short analysis time.	Wide concentration range not covered. Requires complimentary techniques for confirmation. Lack of external validity of the method.
8.	[70]	NMR	Method for the determination of paraffins, olefins, naphthenes and aromatics in gasoline.	Limited concentration range for studied classes. Lack of external validity of the method.
9.	[71]	NMR	Method for determination of aromatic, olefinic and aliphatic fractions including benzene and methyl tert-butyl ether in gasoline. Total analysis time per sample is less than 20 min.	Concentration range not mentioned specifically for the studied classes. Lack of external validity of the method.
10.	ASTM D5986	GC–FTIR	Standard method for determination of gasoline boiling range aromatic and oxygenated adulterants.	Applicable to the mentioned ranges i.e. 0.1%–2.0% benzene, 1.0%–15.0% toluene, 10.0%–40.0% total (C ₆ –C ₁₂) aromatics and 1.0%–20.0% for ethers and alcohols (in vol%).
11.	[21]	GC×GC with multivariate analysis	Method for detection of adulteration in commercial gasoline. Calibration blends of gasoline and white spirit, kerosene and paint thinner.	Lack of external validity of the method.
12.	[52]	GC×GC	Method for determination of benzene and total aromatics in gasoline. Complies to the concentration ranges provided in ASTM D5769 and D5580.	Low sensitivity for adulterants. Lack of external validity of the method.
13.	ASTM D6591	HPLC	Method for determination of mono-, di-, tri- and PAHs in diesel. Good linearity range of 0–26.0% for PAHs.	Compounds containing sulfur, nitrogen, oxygen and conjugated di- and poly-alkenes, if present, are identified as possible interferents. FAMES are also reported as interferents.
14.	[25]	HPLC with multivariate analysis	Method for determination of triacylglycerols (vegetable oils and fats) in diesel.	Lack of external validity of the method.
15.	ASTM D5186	SFC	Standard method for the determination of aromatics and PAHs content in diesel.	Used as alternative method in conjunction with other sensitive methods.
16.	[23]	NMR	Method for detection and quantification of kerosene adulteration in diesel.	Lack of external validity of the method.
17.	[74]	NMR	Method for estimation of total aromatics and their distribution in diesel.	Lack of external validity of the method.
18.	[59]	GC×GC	Method correlates very well with the standard IP-391. Method for estimation of alkanes, naphthenes, mono-, di-, tri-aromatics and fatty acid esters in diesel/bio-diesel blend.	Determination of vegetable oils in diesel not possible. Lack of external validity of the method.
19.	[60]	GC-FIMS	Method for determination of detailed hydrocarbon types in diesel fuel. The method correlates very well with ASTM D2786 and ASTM D3239.	Lack of external validity of the method.
20.	[63]	SFS	Method for determination of kerosene in gasoline and diesel. The method linearity is 1–90% (v/v) and superior precision and sensitivity than other SFS based methods.	Lack of external validity of the method.
21.	[26]	SFS	Method for determination of non-transesterified residual vegetable oil in diesel + biodiesel blend. Superior linearity (0.5–25%), precision and sensitivity than other SFS based methods for diesel.	Higher concentration range is not covered along with lack of external validity of the method.
22.	[65]	LED spectro-fluorimetry	Method for determination of non-transesterified residual cooking oil and biodiesel in diesel. Method has application in the range of 2–20% and 2–45% respectively for the above analytes.	Broad concentration range not applicable along with lack of external validity of the method.

ASTM D2789 is the standard test method for low-resolution mass spectrometry based hydrocarbon type analysis of gasoline (up to the boiling range of 210°C). For middle distillates which cover kerosene and diesel fuel, standard test methods ASTM D2425, D2786, and D3239 are applicable. These methods have wide acceptability for quality control of these streams as well as assessing the variations, if any, due to adulteration. Hydrocarbon type distribution into 22 classes for middle distillates is also accomplished through a non-standard method (HC22) internally validated in our laboratory. This method requires mass resolving power at around 10,000 and differentiates overlapping accurate mass peaks arising from different hydrocarbon groups [61].

2.4.2. Emission/fluorescence spectroscopy

In this form of spectroscopic analysis, emission or excitation spectra is recorded and the concentration of the analyte is measured which is directly proportional to the intensity of emission. It involves a beam of light acting as a source which causes excitation of molecules to emit visible light. Measurement generally involves the recording of emission spectrum by scanning the emission monochromator while keeping the excitation monochromator fixed. Conversely, the excitation spectrum is obtained by scanning the excitation monochromator keeping the emission monochromator constant at a particular wavelength. However, in the synchronous fluorescence scan or spectroscopy (SFS), both the monochromators are scanned simultaneously leading to spectral simplification, bandwidth narrowing, reduced interferences and enhanced resolution. The SFS is an efficient technique to characterize petroleum liquids such as fuels and lubricants [62]. For adulteration studies, commercial automotive fuels (both gasoline and diesel) collected from Pune, India were analyzed and quantified for adulterants with SFS. The linearity of calibration has been claimed between 1 and 90% v/v and validity of results has been assured. The LOD of 1% has been claimed to be sensitive enough to detect the kerosene as an adulterant in these fuels [63]. A report pertaining to the determination of adulteration of kerosene in gasoline and diesel using SFS followed the fingerprinting of these fuels according to the PAH content [64]. SFS analysis along with multivariate calibration has been used to quantify non-transesterified residual vegetable oil as an adulterant in diesel mean relative error approximately being 4% and its range was reported between 0.5 and 25% (w/w) [26].

LED spectrofluorimetry has also been demonstrated for the analysis of biodiesel in diesel and non-transesterified residual cooking oil in the diesel. The LOD, R-Squared, and limit of quantification (LOQ) for biodiesel were reported to be 3%, 0.9962 and 9.9% respectively. Similarly, these values were 2%, 0.9872 and 7.9% respectively in the residual cooking oil [65]. A study on diesel fuel contamination by adulterants such as kerosene, PDS kerosene, hexane, cyclohexane, turpentine oil, etc. was carried out by Patra et al. by excitation emission matrix fluorescence. The spectra were obtained by subtracting the spectrum of adulterated sample from the reference sample. The authors have claimed the linearity in the range of 0–90% [66]. Similar work has also been reported through the same technique that demonstrated the quantitative determination of kerosene contamination in diesel fuel [67]. Application of flame emission spectroscopy has also been shown in this area which uses the intensity of emitted light from a flame, plasma, and spark at a particular wavelength. Paulo et al. classified gasoline adulteration with different adulterants like kerosene, rubber solvent, thinner, turpentine, and ethanol using flame emission spectroscopy along with chemometric approach. In this method, classification of adulterated samples with different solvents is mainly due to the distribution of radicals during the burning of gasoline samples [68].

2.4.3. Nuclear magnetic resonance spectroscopy

Nuclear magnetic resonance spectroscopy (NMR) has been immensely utilized for characterization and adulteration of petroleum fuels and reviewed for its relevance in characterizing crude oils and their mixtures [69]. Usually, ASTM D5292 is a standard test method for determination of aromatic carbon contents of petroleum derived oils by high-resolution NMR spectroscopy. ASTM D4808 is employed for calculation of hydrogen content of light and middle distillates, gas oils, and residues through low-resolution NMR. These standards do not specifically address the adulteration issues but considered vital in providing the compositional clues and useful in complementing results obtained with other analytical techniques. For hydrocarbon type analysis of gasoline, studies have been reported that propose a fast and efficient method based on ^1H NMR [70,71]. NMR along with other non-destructive techniques plays a significant role in the analysis of samples without much pretreatment. From the adulteration point of view, in addition to the typical hydrocarbon classes, other compounds such as olefins, oxygenates and benzene content determination in gasoline with more than 97% confidence is reported [8]. Multivariate analysis of ^1H NMR data to identify the presence of organic solvents and heavy aliphatic hydrocarbons in gasoline at concentrations as low as 5% has been proposed [5]. A study involving the determination of viscosity, API gravity, acid number and the refractive index of light, medium and heavy petroleum fractions through ^1H NMR and comparison of results with ASTM methods was carried out. On the basis of comparison, a correlation between proton relaxation time and physico-chemical properties has been reported which is proposed as a basis for the routine analysis of petroleum fractions and their quality control [72].

^1H NMR data has been used for discriminating between adulterated and unadulterated gasoline samples. The gasoline samples differed in octane number and their NMR data with multivariate analysis provided valuable clues [73]. The ^1H NMR spectral analysis along with multivariate statistical technique was used for quality control of gasoline samples which involved their fingerprinting and pattern-recognition [7]. Bansal et al. had proposed ^1H NMR method for determining diesel adulteration by estimating distribution in aromatic hydrocarbon classes such as PAHs. The method is reported to yield results in close agreement with standard test method IP-391 [74]. In a separate method, NMR time-domain methodology was developed for the quantification of kerosene adulteration in diesel. The basis adopted was differences in the transverse relaxation times obtained separately for diesel, selected adulterant and the adulterated samples [23].

2.4.4. Infrared/Raman spectroscopy

Fourier transform infrared spectroscopy (FTIR) provides valuable functional group information about the sample. It has been successfully demonstrated to detect the adulteration in gasoline and diesel fuels. A method for the direct determination of benzene in gasoline samples based on flow-injection FTIR is reported with RSD values less than 10% [75]. Multivariate analysis involving FTIR has emerged as a sensitive method to detect adulteration of gasoline. Various multivariate classification methods have been presented and compared as a part of the validation of results [76]. In several cases near infrared (NIR) spectral region have been chosen [22,77]. Pereira et al. proposed an FTIR-based method for gasoline adulteration with the help of multivariate analysis. The results indicated distinct discrimination between the adulterants with close similarity as that of gasoline at the minimum concentration of 8% [22]. Analysis of petroleum-based fuels with FTIR and chemometrics has been reviewed which encompasses important literature references in this area and discussed the application of various analytical approaches, quality control and adulteration [78]. A

study carried out by Diehl et al. utilized hyphenated analytical technique, GC–FTIR which is used less often in this area. The results were extracted from the full IR spectra and reportedly comparable with GC–MS. The authors also claimed that GC–FTIR was capable of simultaneously measure oxygenates along with the aromatic compounds in contrast to GC–MS. However, since the GC is common to both these hyphenated techniques, the comment appears invalid as the sensitivity of MS is unusually high unlike FTIR [79].

Measurement of diesel adulterants through FTIR has been investigated by many researchers. Diesel and its blend with bio-diesel adulterated with raw vegetable oil have been analyzed by FTIR [80]. De Souza et al. determined adulteration of diesel with residual automotive lubricants by utilizing statistical data analysis tools. The range of adulterant determined through FTIR remained within 1.75–30.0% [81]. The combination of vibrational spectroscopic techniques such as FTIR and Raman combined with multivariate analysis has been used as a powerful tool for the analysis of petroleum fuels. Marinovic et al. developed an analytical method for quality assessment and monitoring of diesel adulteration through Raman and infrared spectroscopy along with statistical methods. It revealed important properties of diesel such as density, viscosity, distillation recovery, cetane number etc. [82]. FTIR has also been employed to determine bio-diesel content (during a trans-esterification reaction) and other adulterants such as vegetable oil, residual oil from cooking, residual lubricating oil etc. in diesel. In this method, oil content and bio-diesel content in the mixture of petro-diesel/bio-diesel and vegetable oil have been reported with a high accuracy range of 95.32%–99.99% for varying composition of fuel and adulterants [83].

Like other analytical techniques, Raman spectroscopy also finds an important position in the variety of applications for establishing vibrational, rotational, and other low-frequency modes in a chemical system. In view of its simplicity, it has been used in the analysis of aromatics and olefins in the hydrocarbon mixtures [69]. Determination of prominent fuel properties through Raman spectroscopy has been carried out for gasoline. The analysis involved utilization of Raman spectra in the quantitative analysis of gasoline and predicting RON. The classification and branding of gasoline from various refineries in China is achieved through Raman spectroscopy [84]. In specific cases, Raman spectroscopy provides higher spectral detail and accuracy as compared to FTIR. Adulteration of commercial and spiked gasoline have been investigated with Raman spectroscopy with a limit of detection below 5% [85] in addition to the determination of RON, MON and Reid vapor pressure of a large number of commercial petroleum fuel blends by FT-Raman spectroscopy [86].

2.4.5. Other methods

Turbidimetry based phase-titration method for the determination of kerosene adulteration in gasoline was investigated in 1990 by Bahari et al. [87]. The phase-titration method was first proposed for the analysis of binary petroleum mixtures involving estimation of adulteration of gasoline and diesel with kerosene for a range of 0–30%. In this method, distilled water was used as a titrant, acetone as an additive to homogenize water and hydrocarbons and the end point was turbid. No instrumentation and portability of the method were highlighted as the merit of this study [88]. A method for the detection of adulterants (kerosene and diesel) in gasoline using ultrasonic radiations was proposed which used the measurement of the speed of ultrasonic waves in the samples using pulse echo method; however, the validation procedure is missing [89]. Photo-thermal gas analyzer method for monitoring gasoline adulteration consisted of measuring the time-evolution of gasoline vapors in an air-filled closed cell. In this case, the thermal diffusivity of the

gasoline vapor/air was used for detecting the gasoline adulteration with good accuracy [34]. Thermal methods such as differential thermal analysis, differential scanning calorimetry and thermogravimetry have also been used for the analysis of varied petroleum products like fuel, lubricating oils, greases, paraffin waxes and asphalts [90]. Fonseca et al. have evaluated the thermal behavior of gasoline doped with kerosene using thermogravimetric technique [91]. A simple method for estimating high boiling hydrocarbons (kerosene/diesel) in motor gasoline has been reported based on the gravimetric determination of the residue after evaporation [92].

3. Conclusion

Adulteration of petroleum fuels is widespread in the developing countries and strongly manifested through the deterioration of the environment and adverse effect on its citizen's health. In view of the targeted implementation of Euro-VI emission norms in coming years, the monitoring of adulteration in transport fuels would be highly challenging. The standard methods pertaining to QA/QC of gasoline and diesel fuels are of great help in ascertaining the deviations from their standard specifications. In addition, potential laboratory developed methods for adulteration screening and determination are of great importance provided they are externally validated. The major thrust is on the evaluation of physico-chemical properties which indicate adulteration at significant levels, however, in the changing scenario, low levels of adulteration monitoring rely greatly on characterization at molecular scales. The offsite analysis is often multi-step, complicated, time taking and may involve sophisticated instruments but considered essential. As a result, chromatography and spectroscopic methods shall continue to play a significant role in addressing this concern, especially in offsite verification.

This review article has discussed some of the important literature reports which involved diverse analytical approaches for monitoring of adulteration of petroleum fuels. The adopted approaches were based on important physico-chemical properties and their relation with adulteration. Besides, the role of chromatographic and spectroscopic techniques has also been discussed. GC, HPLC, GC–MS, GC×GC, FTIR, Raman and NMR have provided a significant contribution in this area. Chemometrics and multivariate methods of analysis have demonstrated a tremendous potential in differentiating signals from samples doped with variable amounts of adulterants [35,39]. In addition to other techniques, NMR and FTIR showed excellent correlation between the studied variables (R-squared) and precision exhibited by RSD [76,77,79,84]. In several discussed reports, GC, Raman, and FTIR have been used for the determination of RON/MON and Reid vapor pressure of the samples, however, in the absence of validation procedures (inter-laboratory comparisons or proficiency testing programs), they may have limited correlation with physico-chemical results. If validated, such 'green' methods can be substituted for physico-chemical determinations with minimum sample requirements.

In most of the analytical methods, sensitivity, selectivity, robustness, accuracy, and precision remained essential factors that governed their popularity and efficiency. In comparing the overall implications of the threat posed by adulteration, the green aspects of the adopted analytical methods are largely related to the overall sample quantity utilized during the assay. The physico-chemical methods utilize the higher amount of samples as compared to the chromatographic/spectroscopic methods where, except dilution, no other form of sample pre-treatment is required in several cases. The massive use of advanced statistical data treatments has further facilitated the solvent-free approach with high accuracy and better resolution of data [93].

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.trac.2017.04.011>.

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